

The background of the slide is a 3D grid of white cubes. One cube, located in the upper right quadrant, is red and stands taller than the others, creating a focal point. On the left side, there is a red rounded rectangular overlay containing white text.

Object Oriented Analysis and Design ICT2132

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LECTURE 01:

**OBJECT ORIENTED ANALYSIS
AND DESIGN BASIC CONCEPTS**



Objectives

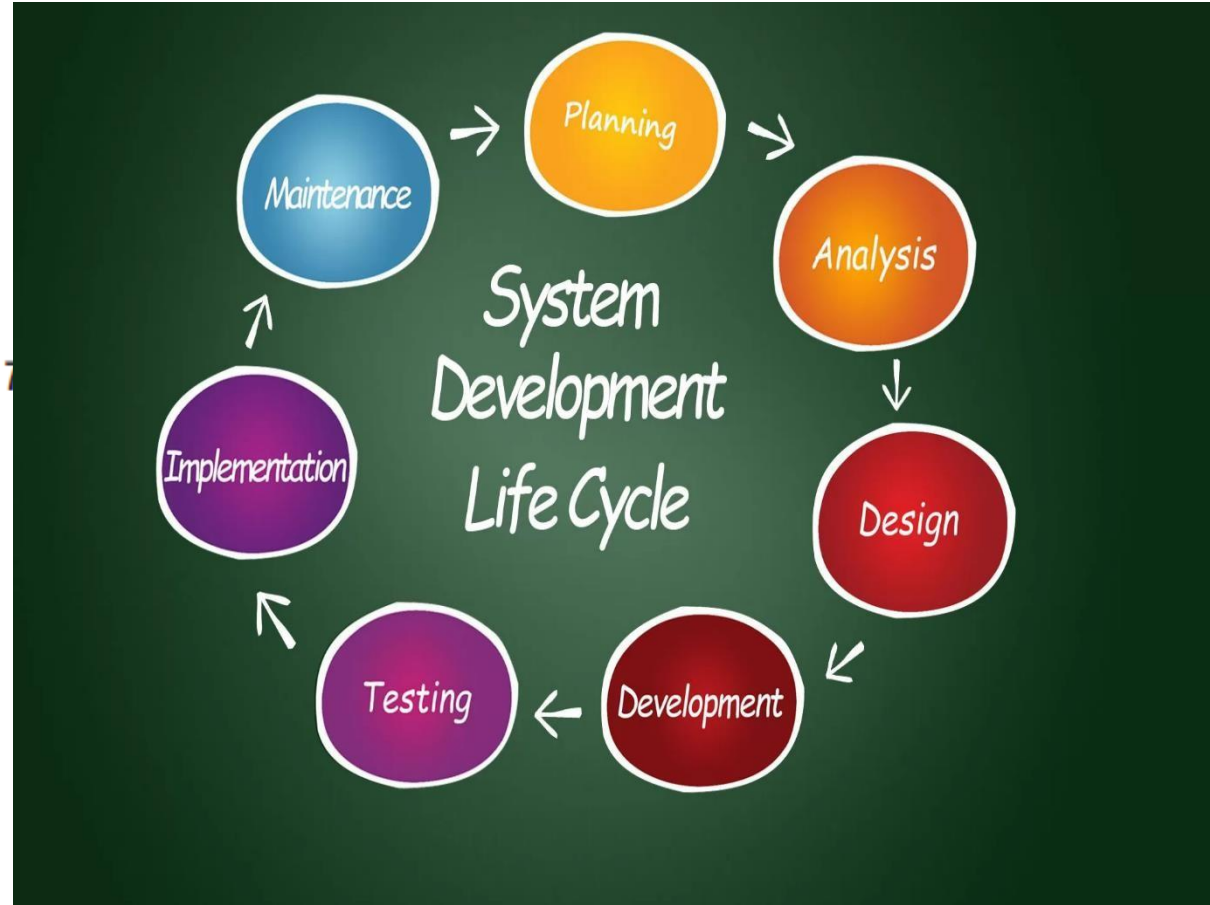
After successful completion of this lecture, students should be able to:

- Identify what is object orientation
- Describe the principles of object orientation



The background of the image is a 3D grid of white cubes, creating a textured, isometric effect. One cube, located in the upper right quadrant, is colored red and stands slightly taller than the others, drawing attention to the title.

System Development Life Cycle



What is Object Orientation?



- 
- Object orientation is about viewing and modeling any system as a set of interacting & interrelated objects.
 - Object-Oriented (OO) systems development is a way to develop software by building self-contained modules that can be more easily:
 - Replaced
 - Modified
 - and Reused
 - In an object-oriented environment, software is a collection of discrete objects that encapsulates their data as well as their functionality to model real world objects.

Object Orientation – Cont...

OOA – Object Oriented Analysis

- Emphasize finding and describing the objects.
(or concepts) in the problem domain, i.e: domain objects.

OOD - Object Oriented Design

- Emphasize defining software objects and how they collaborate to fulfill the requirements.

OOP - Object Oriented Programming (implementation)

- Designed objects are implemented in a programming language.
- Implementation is also known as *Coding* or *Construction*.



Object Orientation – Cont...

- OO development offers a different model from the traditional software development approach.
- Functions (**behaviour**) and data (**state**) relating to a single object are self-contained or encapsulated in one place.
- Objects are grouped into classes
 - In object-oriented terms, we discover and describe classes involved in the problem domain.



Why Object Orientation?



- 
- Simplicity
 - Reusability
 - Increased quality
 - Faster development
 - Easily maintainable
 - Scalable
 - Modularity
 - Modifiability

Exercise

Write down the disadvantages of object oriented approach

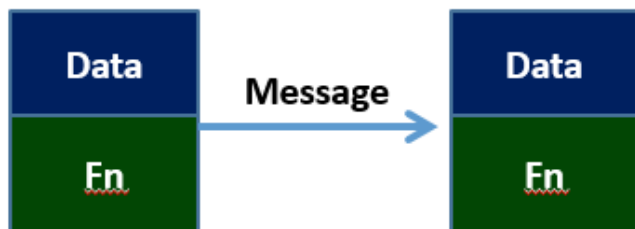


OO Paradigm vs Traditional Procedural Paradigm



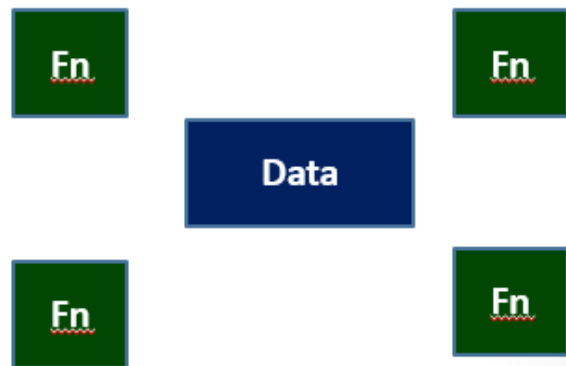
Object Oriented

- Collect data & functions into a class



Traditional

- Procedures (functions) & data are designed separately





Object Oriented Programming

Program is divided into small parts called **objects**

Supports inheritance: more code reusability

Function overloading is supported

Access specifiers are supported

More secure as it provides data hiding

Data is more important than function

Examples: C++, Java, Python, C# etc.

Procedural Programming

Program is divided into small parts called **functions**

No inheritance: limited code reusability

Function overloading is not supported

Access specifiers are not supported

Less secure as it does not provide a proper mechanism to hide data

Function is more important than data

Examples: C++, C, FORTRAN, Pascal, Basic etc.

OO Environment

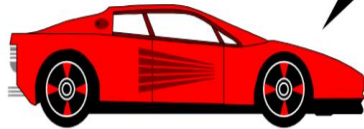


What is an object?

- The term object was first formally utilized in the Simula language to simulate some aspect of reality.
- An object is a real world entity.
 - It knows things (has **attributes**).
 - It does things (provides services or has **methods**).



Attributes



I am a Car.
I know my color,
manufacturer, cost,
owner and model.

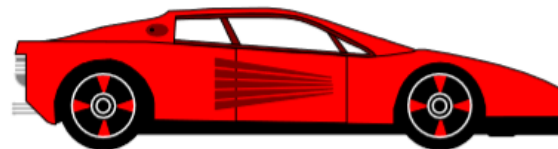


I am a Fish.
I know my date of
arrival and
expiration.

Methods



I know how to
compute
my payroll.



I know how
to stop.

A red 3D cube is positioned on a white grid of squares, which is part of a larger 3D structure. The cube is bright red and has a glossy finish. The grid is composed of white squares, and the overall scene is set against a light gray background.

Object's Attributes and Methods

➤ Attributes

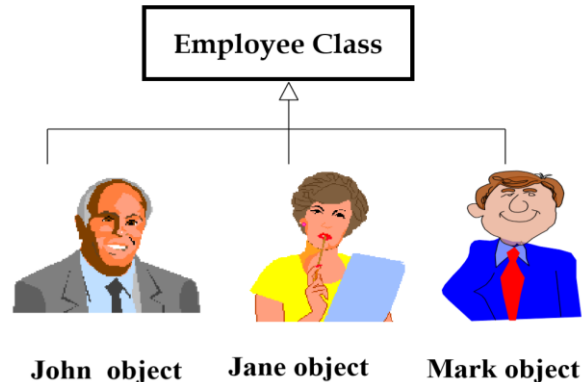
- Represented by data type.
- They describe objects states.
- In the Car example the car's attributes are:
 - color, manufacturer, cost, owner, model, etc.

➤ Methods

- Define objects behavior and specify the way in which an Object's data are manipulated.
- In the Car example the car's methods are:
 - drive it, lock it, tow it, carry passenger in it.

Objects are Grouped into Classes

- A class represents **a collection of objects having the same characteristic properties that exhibit.**
- The class **employee**, for example, defines the property **name**.
- Each individual employee (object) will have a value for this property, such as “John,” “Jane” or “Mark.”



OO Analysis

- Examines requirements from the perspective of the classes and objects found in the vocabulary of the problem domain.
- Emphasis is on finding objects or concepts in the problem domain.

Ex: Library info system

- Concepts: Book, Library,.....





OO Design

- Structures are developed whereby sets of objects collaborate to provide the behaviours that satisfy the requirements of the problem.
- Emphasis is on defining SW objects & how they collaborate to fulfill the requirements.

Ex: In the library system a book object may have a **title** attribute and a **getChapter** method.



OO Design

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Ex: In the library system a book object may have a **title** attribute and a **getChapter** method.

A 3D perspective view of a grid of white cubes. One cube in the upper right quadrant is colored red, standing out from the rest of the white cubes. The text "Principles of Object Orientation" is centered over the grid.

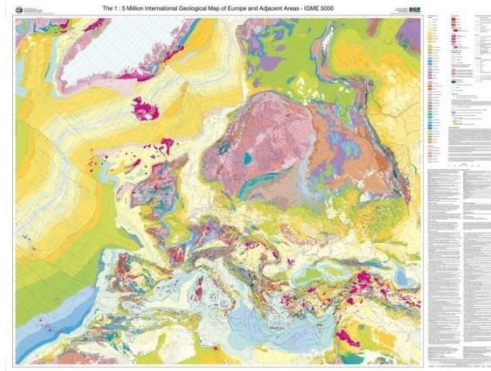
Principles of Object Orientation

- 
- A red cube is positioned on a gray grid of squares, which is part of a larger 3D environment with a light gray background.
- **Abstraction**
 - **Encapsulation**
 - **Polymorphism**
 - **Inheritance**

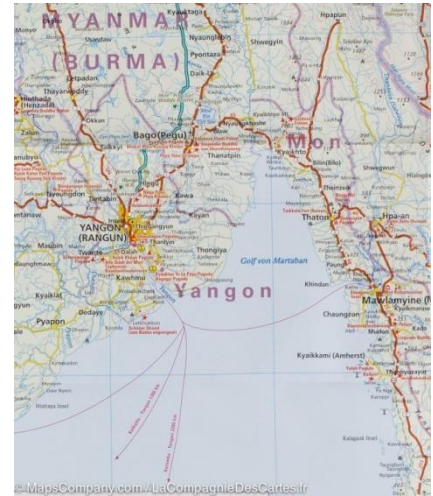
Abstraction

- A form of representation that includes only what is important or interesting from a particular viewpoint.
- It **includes most important aspects** of a given system while **ignoring less important details**.

Ex: A **map** is an abstract representation



Geological map



Road map

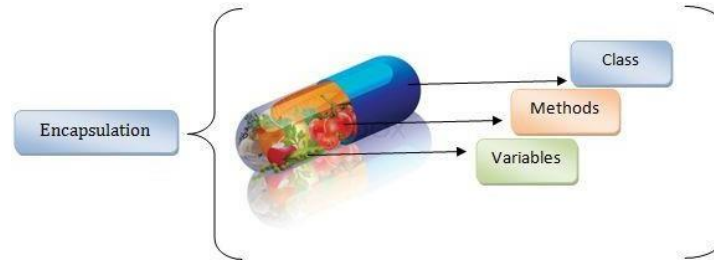
Abstraction Example



- If you look at the image, you can see when you get a call, we get an option to either pick it up or just reject it.
- But in reality, there is a lot of code that runs in the background.
- So here, you don't know the internal processing of how a call is generated.

Encapsulation and Data Hiding

- **Encapsulation** - Packaging related data and operations together.
- Information hiding is a principle of hiding internal data and procedures of an object.
- Separates the external aspects of an object from the internal implementation details of the object, which are hidden from other objects.

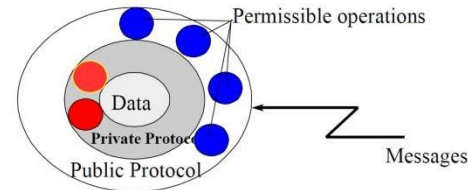


Encapsulation cont...

- Typically, the structure of an object is hidden, as well as the implementation of its methods.

Advantages

- Ensuring data integrity
 - Access to the encapsulated data is limited to the operations defined on the data.
- Flexibility to change implementation of an object without affecting its clients



Encapsulation Example

- Consider a coffee shop where the customer, waiter & kitchen are three entities.
- For the customer it doesn't matter how the kitchen functions.



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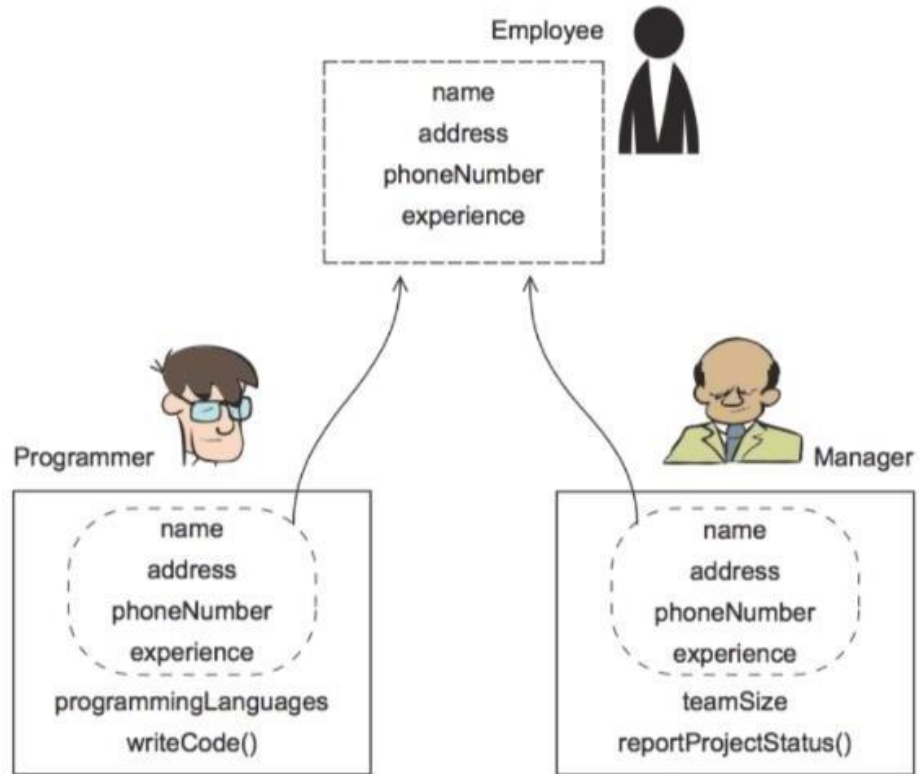


Inheritance

- Inheritance is a relationship between classes where one class is the parent class of another (derived) class.
- Inheritance allows classes to share and reuse behaviors and attributes.
- The real advantage of inheritance is that we can build upon what we already have and, reuse what we already have.



Inheritance Example



Polymorphism

- The word polymorphism means having many forms.
- The word “**poly**” means many and “**morphs**” means forms, So it means many forms.
- In simple words, we can define polymorphism as the ability of a message to be displayed in more than one form.
- Polymorphism allows us to perform a single action in different ways.



Polymorphism Example

- A person at the same time can have different characteristic.
- Like a man at the same time is:
 - a father
 - a husband
 - an employee
- So the same person posses different behavior in different situations.
- This is called polymorphism.



THANK YOU!

